Name: _____ Class and student number: _____

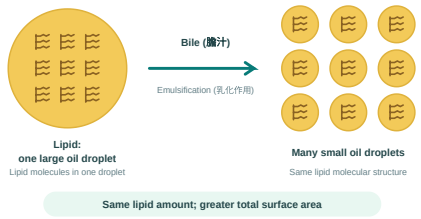
Digestion of lipids and fair comparisons

Lipids in food must be digested into small molecules before absorption.
Four sets compare adding X, Y or both to an oil–water mixture.
Two further tests compare boiling X or Y and then cooling it.
Use the notes and figures to complete B, then check with C. Keep A available.

Quick notes (use while answering)

X: bile (膽汁)	Made in the liver, stored in the gall bladder (膽囊); contains no digestive enzymes.
Emulsification (乳化作用): physical digestion (物理消化)	Large oil droplets become small droplets. Lipid molecules stay unchanged; total surface area increases, helping Y act.
Y: lipase (脂肪酶)	Catalyses hydrolysis that changes chemical bonds: chemical digestion (化學消化). Simplified complete hydrolysis: triglyceride (甘油三酯) + $3H_2O$ → glycerol (甘油) + 3 fatty acids (脂肪酸). The enzyme is not used up.
Heat and denaturation	Lipase (脂肪酶) is a protein. Boiling can change its shape and stop it working (denaturation). Cooling does not restore it in this model. Compare at 37°C.
Appearance and product evidence	Layers, a thinner oil layer, cloudiness or clarity alone cannot prove glycerol (甘油) and fatty acids (脂肪酸) formed. Chemical tests and repeated tests are needed.
Model and limits	The figures are preset model data, with no speed measurements. Individual juice amounts differ between combined and single-juice sets, so a measured combined effect or faster rate cannot be proved.

01 • Physical digestion (物理消化): emulsification (乳化作用), with no change in molecular structure



Lipid molecules stay the same during emulsification (乳化作用). Not to scale.

> Variables：Independent: changed on purpose. Dependent: observed/measured result. Controlled: kept the same.
> Control：Choose a baseline for the question. A control may contain reagents, for example in a heating comparison.

Example: a testable prediction

With other conditions equal, 2 mL X disperses large oil droplets because bile (膽汁) emulsifies lipids (乳化脂質). Compare water with X to test it. Testable does not mean already supported.

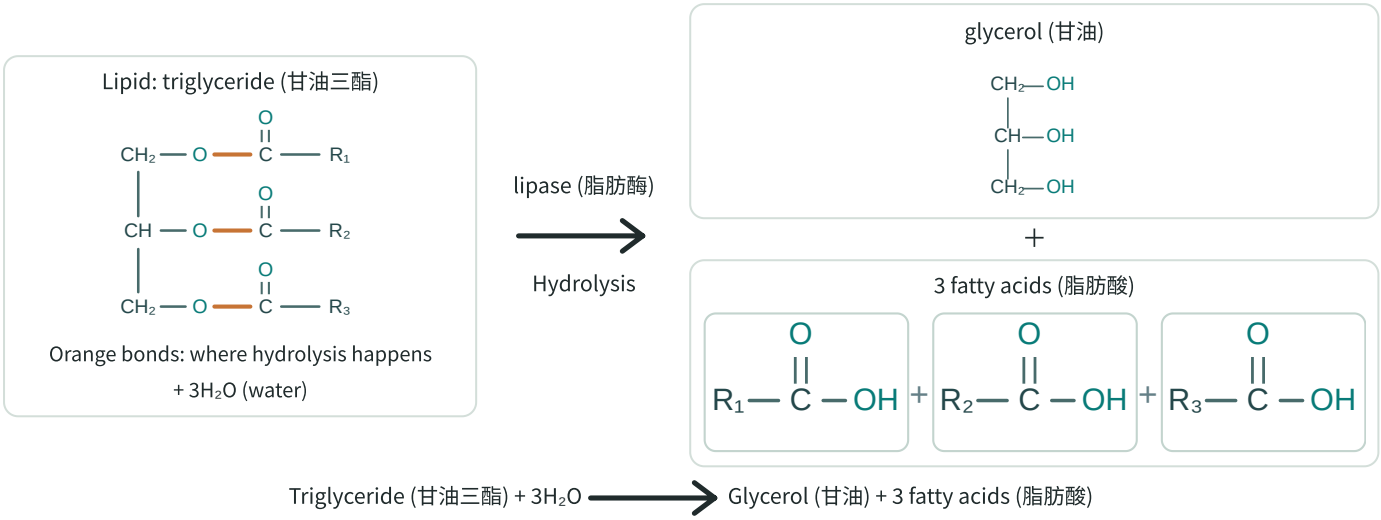
Example: classification and observation

If only shaking changes, it is independent; appearance is dependent, and other conditions stay fixed. “Particles at the bottom” is an observation; “chemically broken down” needs evidence.

The total activity time is 15–20 minutes. Follow the teacher’s time limit for reading, practice, checking and correcting.

Experiment setup and data

02 • Chemical digestion (化學消化): hydrolysis forms different molecules



Complete-hydrolysis model. R_1–R_3 are fatty acid (脂肪酸) chains; orange bonds show where hydrolysis occurs.

Keep starting oil/water amounts, total added volume (2 mL), 37°C, observation time and shaking the same. Added liquids are shown in the figures. Cool boiled liquids to 37°C first.

Figure 1. Four setups and their observed appearances

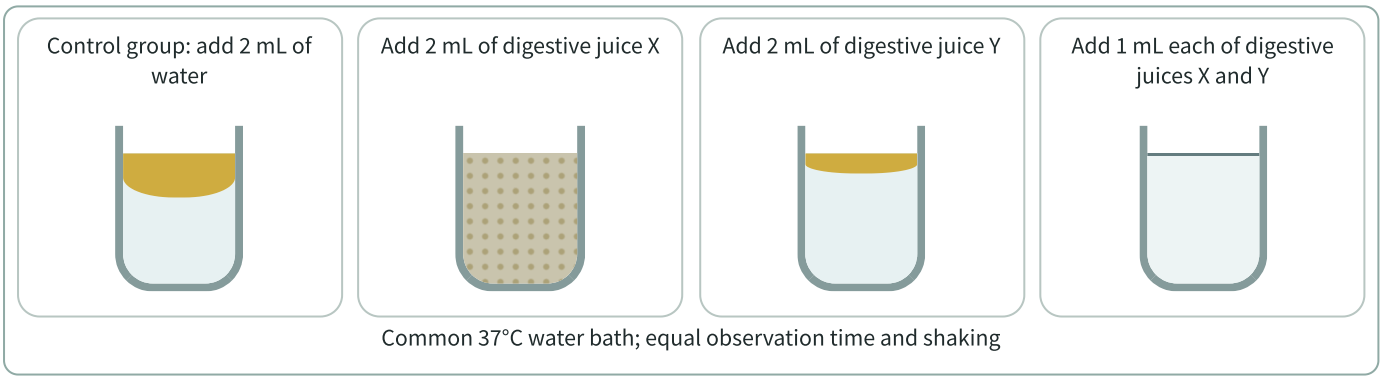
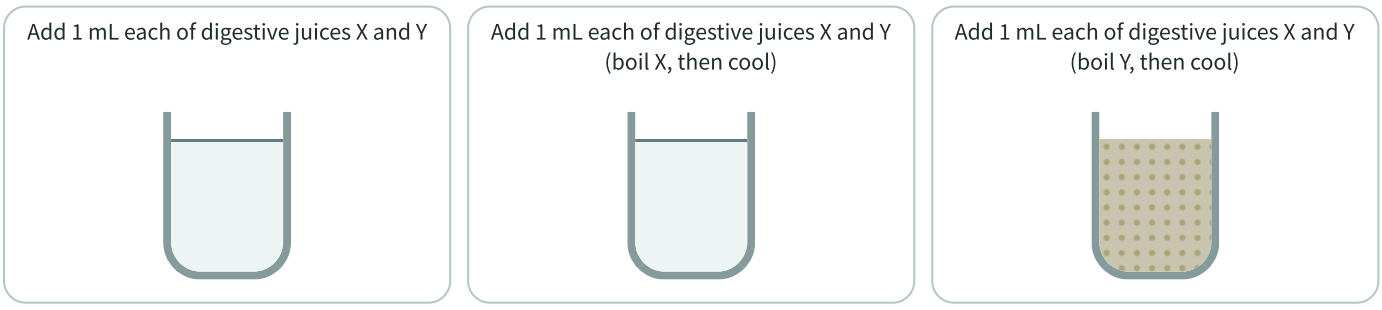


Figure 2. Comparison after boiling and cooling



These images are static data from the current teaching model. Appearance answers are not filled in. They are not direct product-test results. No speed was measured. Record your appearance choices in B, then check with C.